Queen Mary
University of London

EBU5376
Digital Signal Processing

Filters – Frequency-domain analysis

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Outline

- Filter Analysis using Frequency Response
  - Cut-off frequency
- Classification of Filters
  - lowpass, highpass, bandpass, bandstop
  - FIR, IIR

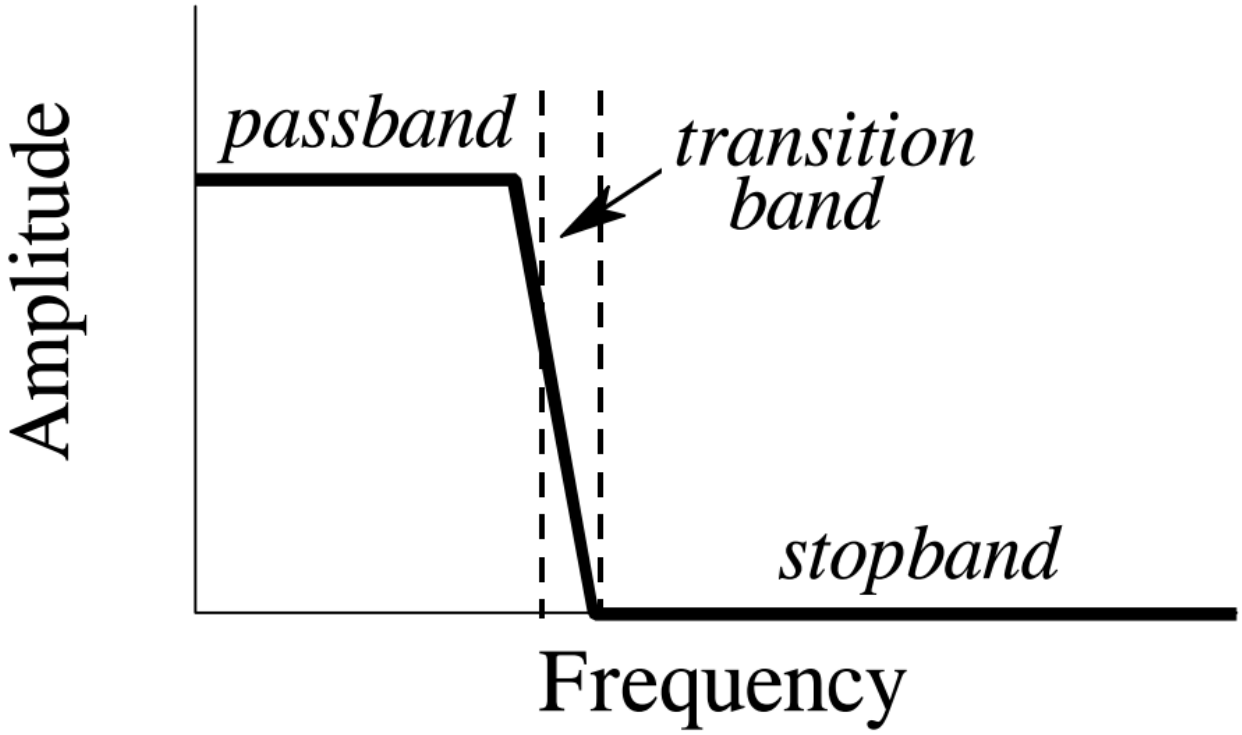
Filter Analysis – Frequency Response

- Both impulse and step responses are time-domain responses
- **Frequency response** shows how the filter changes different frequency components in the input
- Frequency responses can be decomposed into
  - Amplitude Response
    - ✓ shows how the filter changes the amplitude of different frequency components
  - Phase Response
    - ✓ shows how the filter changes the phase of different frequency components

Filter Analysis – Amplitude Response

- By increasing / decreasing the amplitudes of different frequencies, the filter can achieve
  - Feature extraction
  - Noise reduction
  - …
- There are three parameters of amplitude responses:
  1. Passband
  2. Transition band
  3. Stopband

Filter Analysis – Amplitude Response



Filter Analysis – Amplitude Response

Filter analysis using Amplitude Response

- The passband refers to those frequencies that are passed,
- The stopband contains those frequencies that are blocked.
- The transition band is between.

Filter Analysis – Amplitude Response

Filter analysis using Amplitude Response

- The passband refers to those frequencies that are passed,
  - For the passband frequencies to move through the filter unaltered, there must be no passband ripple
- The stopband contains those frequencies that are blocked.
- The transition band is between.

Filter Analysis – Amplitude Response

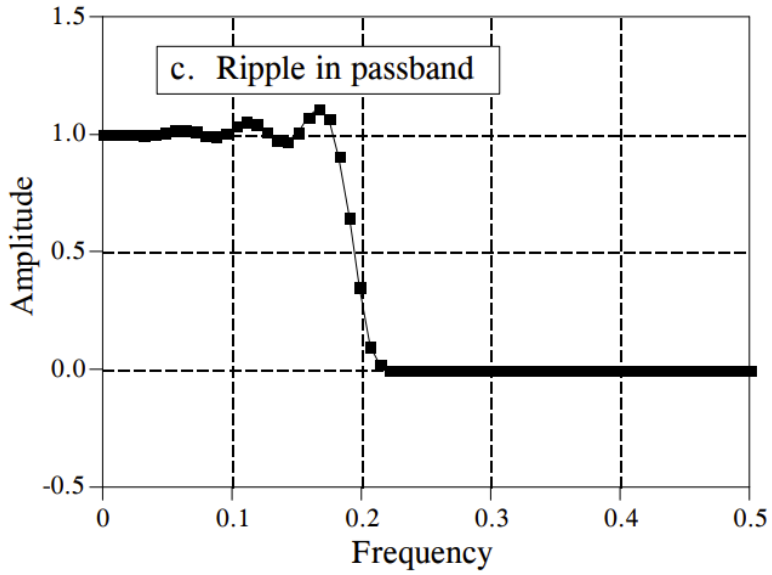
Amplitude Response Analysis – Passband

POOR | GOOD

Filter Analysis – Amplitude Response

Amplitude Response Analysis – Passband

POOR



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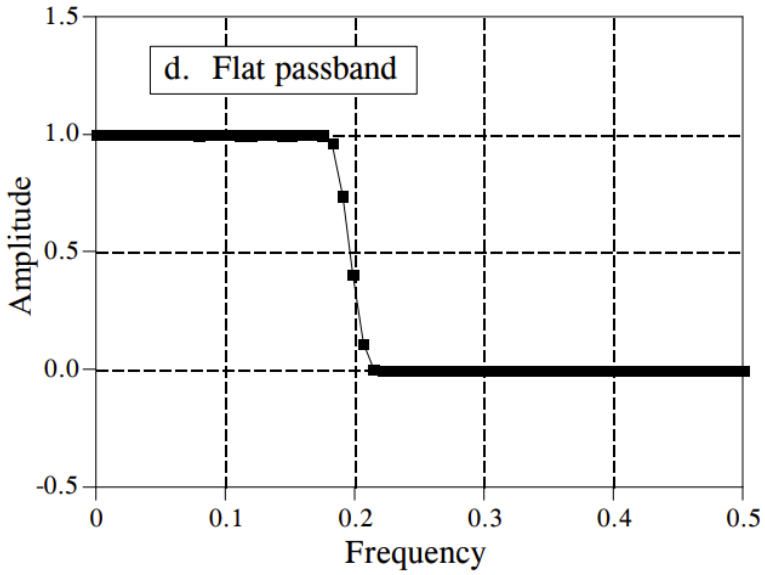

Filter Analysis – Amplitude Response

Filter analysis using Amplitude Response

- The passband refers to those frequencies that are passed,
- The stopband contains those frequencies that are blocked.
- The transition band is between.
  - To separate closely spaced frequencies, the filter must have a fast roll-off, meaning that the transition band is very narrow.

Filter Analysis – Amplitude Response

Amplitude Response Analysis – Transition band

POOR

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Filter Analysis – Amplitude Response

Amplitude Response Analysis – Transition band

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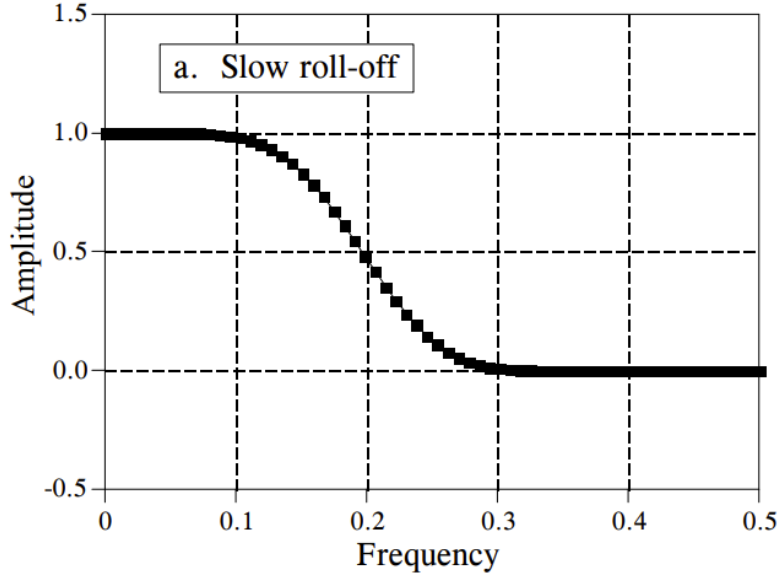

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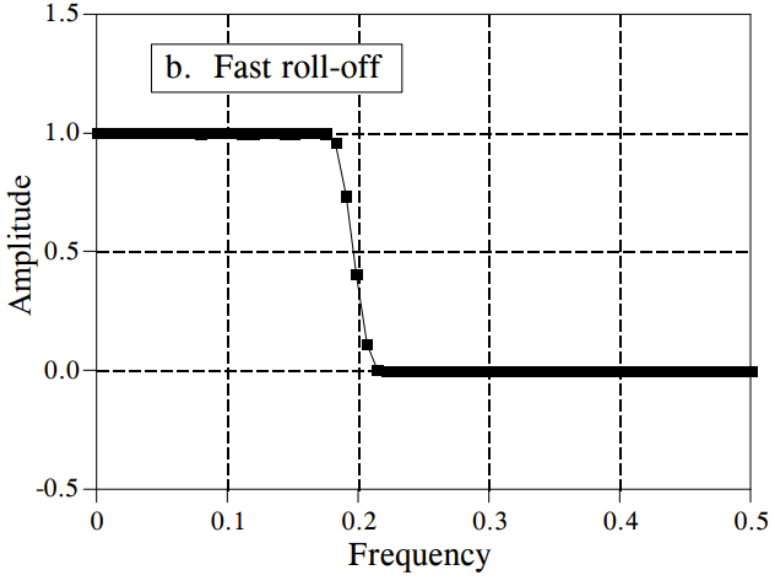


Filter Analysis – Amplitude Response

Filter analysis using Amplitude Response

- The passband refers to those frequencies that are passed,
- The stopband contains those frequencies that are blocked.
  - to adequately block the stopband frequencies, it is necessary to have good stopband attenuation
- The transition band is between.

Filter Analysis – Amplitude Response

Amplitude Response Analysis – Stopband

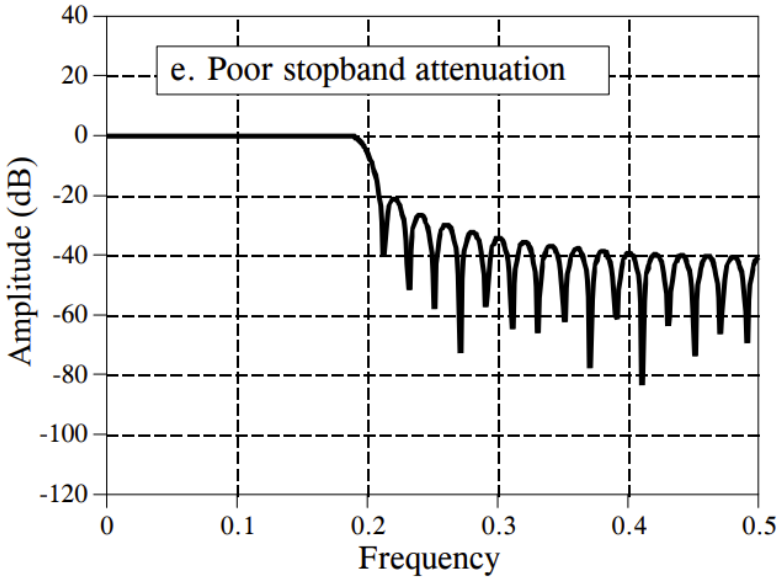
POOR

GOOD

Filter Analysis – Amplitude Response

Amplitude Response Analysis – Stopband

POOR



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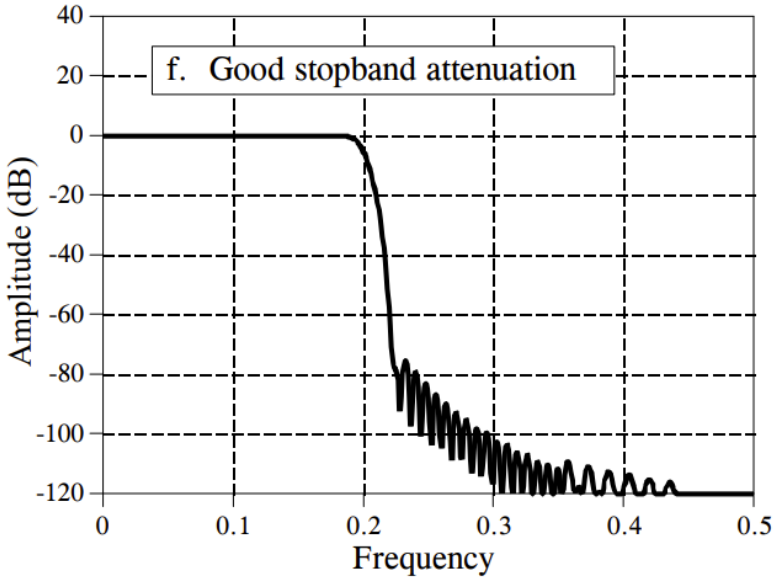


Filter Analysis – Amplitude Response

More on Amplitude Response

- The boundary between the passband and transition band is called the cutoff frequency.
  - In analog filter design, the cutoff frequency is usually defined to be where the amplitude is reduced to from 1 to 0.707 (i.e., -3dB).
  - Digital filters are less standardized, and it is common to see 99%, 90%, 70.7%, and 50% amplitude levels defined to be the cutoff frequency

Filter Representation – MATLAB

$$y[n] + a_1 y[n-1] + \cdots + a_N y[n-N] = b_0 x[n] + b_1 x[n-1] + b_M x[n-M]$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{\sum_{k=0}^{M} b_k z^{-k}}{\sum_{k=0}^{N} a_k z^{-k}}$$

- MATLAB can also help you calculate the filter responses:
  - ‘[h,t] = impz(b,a,n)’ → h is the impulse response
  - ‘[h,t] = stepz(b,a,n)’ → h is the step response
  - ‘[h,w] = freqz(b,a,n)’ → h is the frequency response
- ‘b’ and ‘a’ are the coefficients in the difference equation/transfer function; ‘n’ is the number of samples you want to obtain
- More details in Impulse response, Frequency response, Step response

Filter Representation – MATLAB

- MATLAB Example: Consider the following filter and input

$$x[n] = [0, 1, 1, 2], \qquad n = 0,1,2,3$$

$$y[n] = x[n] - 0.5x[n-1]$$

1. Without using MATLAB, calculate **the impulse response and amplitude response** of this filter
2. Use MATLAB to check your answer to 1.

Filter Representation – MATLAB

- MATLAB Example: Consider the following filter and input

$$x[n] = [0, 1, 1, 2], \qquad n = 0,1,2,3$$

$$y[n] = x[n] - 0.5x[n-1]$$

1. Without using MATLAB, calculate **the impulse response and amplitude response** of this filter
2. Use MATLAB to check your answer to 1.

Answer to 1.

$$h[n] = \delta[n] - 0.5\delta[n-1] = [1, -0.5], \qquad n = 0, 1$$

$$H(\omega) = 1 - 0.5e^{-j\omega} \Rightarrow |H(\omega)| = \sqrt{1.25 - \cos(\omega)}$$

Filter Representation – MATLAB

- MATAB Example: Consider the following filter and input

$$x[n] = [0, 1, 1, 2], \qquad n = 0,1,2,3$$

$$y[n] = x[n] - 0.5x[n-1]$$

1. Without using MATLAB, calculate **the impulse response and amplitude response** of this filter
2. Use MATLAB to check your answer to 1.

```
[h,t]=impz(b,a,20);
h.' % impulse response h[n]
```

Same answer!

```
ans = 1×20
      1.0000   -0.5000         0         0         0         0         0         0 ...
```

```
t.'  % values of n
```

```
ans = 1×20
       0     1     2     3     4     5     6     7     8     9    10    11    12    13 ...
```

Filter Representation – MATLAB

- MATAB Example: Consider the following filter and input

$$x[n] = [0, 1, 1, 2], \qquad n = 0,1,2,3$$

$$y[n] = x[n] - 0.5x[n-1]$$

2. Use MATLAB to check your answer to 1.

```
[hf,w]=freqz(b,a,20);
hf.' % frequency response, H(w)
```

This is the frequency response, we calculate the amplitude response by taking absolute value of it

```
ans = 1×20 complex
     0.5000 + 0.0000i   0.5062 + 0.0782i   0.5245 + 0.1545i   0.5545 + 0.2270i ...
```

```
w.' % angular frequencies, w
```

```
ans = 1×20
         0    0.1571    0.3142    0.4712    0.6283    0.7854    0.9425    1.0996 ...
```

Filter Representation – MATLAB

- MATAB Example: Consider the following filter and input

$$x[n] = [0, 1, 1, 2], \qquad n = 0,1,2,3$$

$$y[n] = x[n] - 0.5x[n-1]$$

2. Use MATLAB to check your answer to 1.

```
abs(hf).'  % amplitude response based on MATLAB result
```

Let's further check our answer via plots

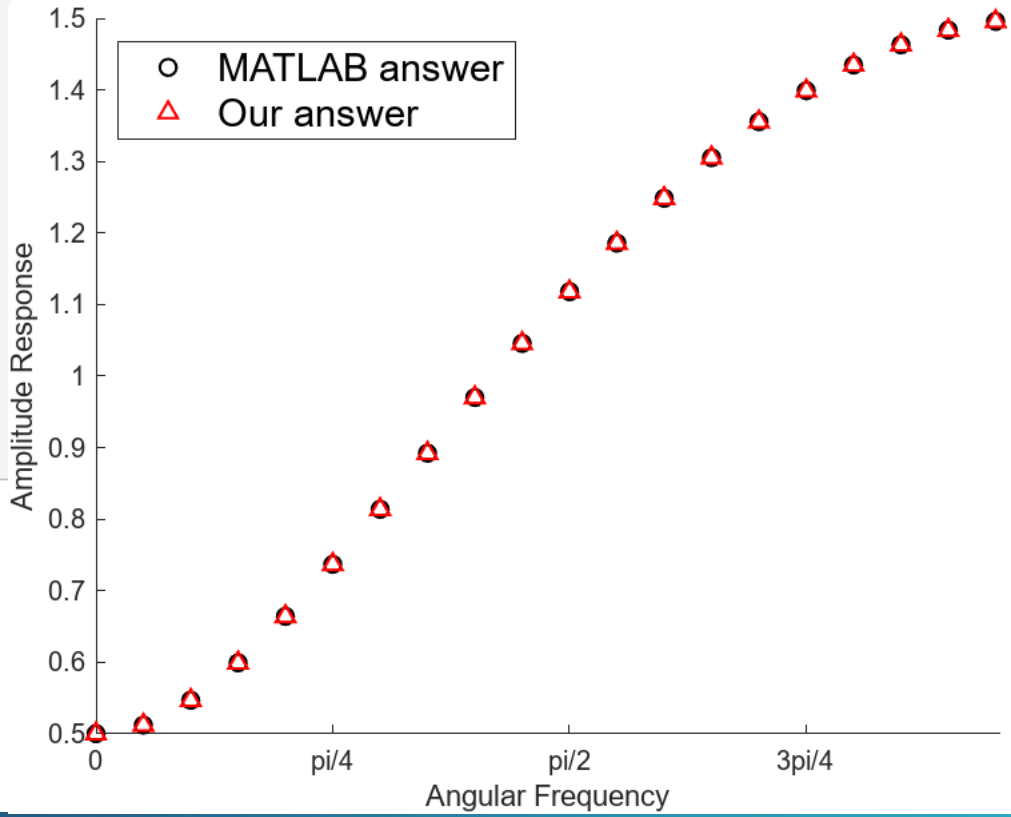
```
ans = 1×20
    0.5000    0.5122    0.5468    0.5992    0.6641    0.7368    0.8138    0.8922 ...
```

```
hf_mag=sqrt(1.25-cos(w)).' % amplitude response that we calculated
```

```
hf_mag = 1×20
    0.5000    0.5122    0.5468    0.5992    0.6641    0.7368    0.8138    0.8922 ...
```

Filter Representation – MATLAB

```
figure()
hold on
plot(w, abs(hf), 'ok',LineWidth=1)
plot(w, hf_mag, '^r',LineWidth=1)
hold off
legend('MATLAB answer','Our answer', ...
    Location='northwest',fontsize=14)
ylabel('Amplitude Response')
xlabel('Angular Frequency')
xticks([0,pi/4,pi/2,pi*3/4,pi])
xticklabels({'0', 'pi/4', 'pi/2', '3pi/4','pi'})
```



Same answer!

Outline

- Filter Analysis using Frequency Response
  - Cut-off frequency
- Classification of Filters
  - lowpass, highpass, bandpass, bandstop
  - FIR, IIR

Filter Classification – Amplitude Response

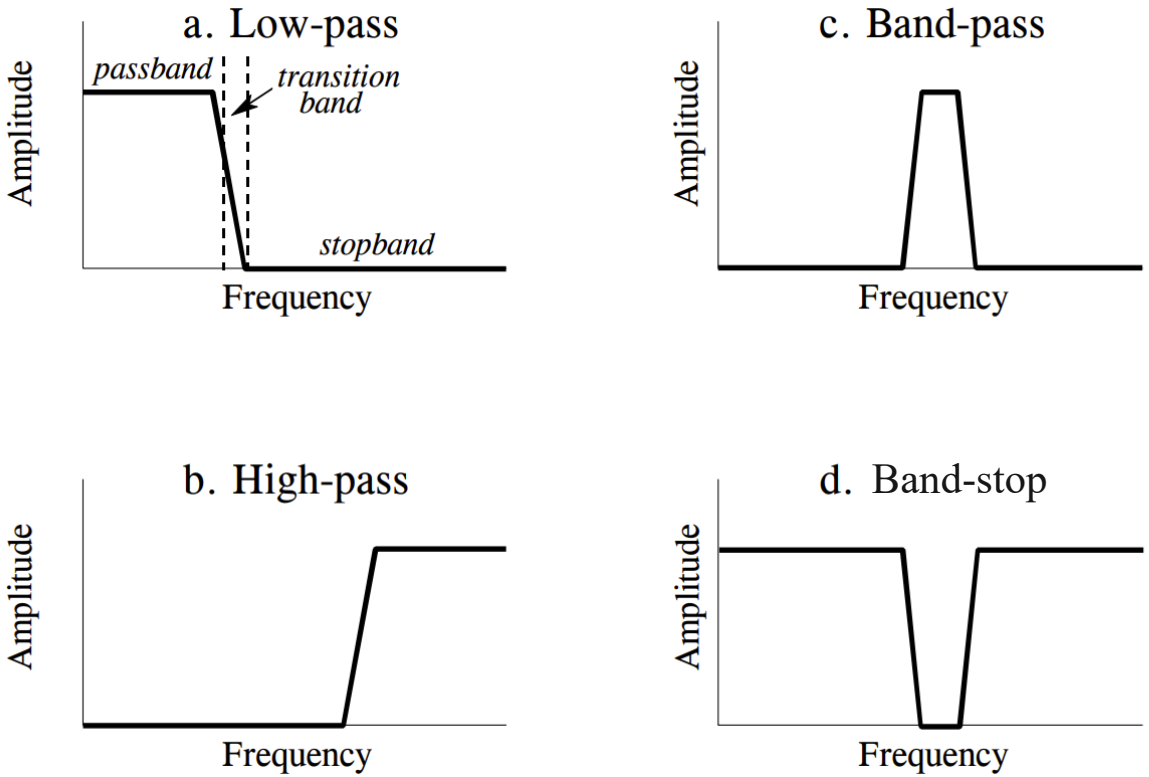
Based on the *Amplitude Response*, filters are classified into:

1. Low-pass filters
2. High-pass filters
3. Band-pass filters
4. Band-stop filters

These are designed by starting with a low-pass filter, and then converting it into the desired response.

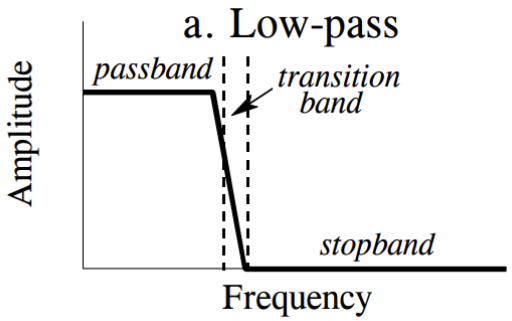
- Hence, most discussions on filter design only give examples of low-pass filters.

Filter Classification – Amplitude Response



Filter Classification – Amplitude Response

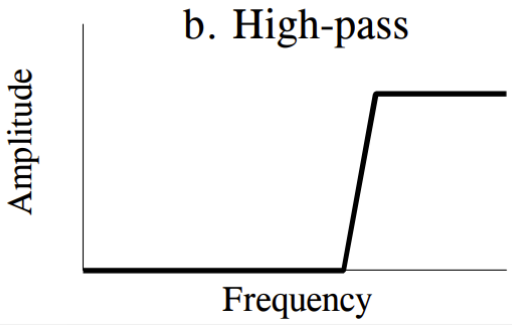
Low-pass filter



- A low-pass filter attenuates high-frequency components of a signal while allowing low-frequency components to pass through.
- It is characterized by a pass-band that includes low-frequency components and a stop-band that includes high-frequency components.
- A low-pass filter is often used to remove noise or unwanted high-frequency components from a signal.

Filter Classification – Amplitude Response

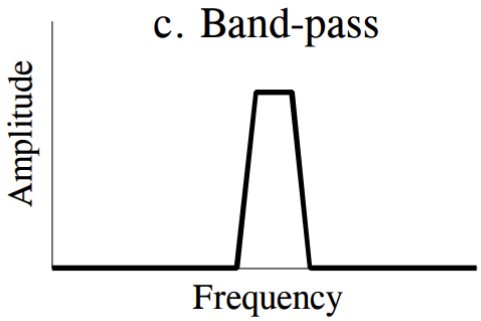
High-pass filter



- A high-pass filter attenuates low-frequency components of a signal while allowing high-frequency components to pass through.
- It is characterized by a pass-band that includes high-frequency components and a stop-band that includes low-frequency components.
- A high-pass filter is often used to remove DC offset or unwanted low-frequency components from a signal

Filter Classification – Amplitude Response

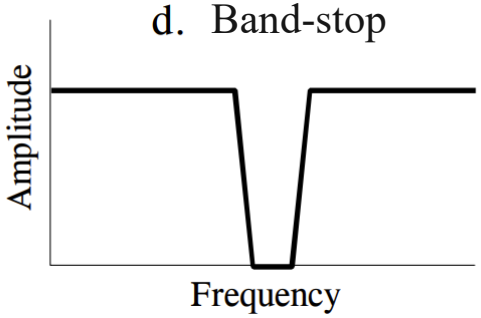
Band-pass filter



- A band-pass filter allows a range of frequencies to pass through while attenuating frequencies outside of that range
- It is characterized by a pass-band that includes a range of frequencies and two stop-bands on either side of the passband.
- A band-pass filter is often used to extract a specific frequency range from a signal or to remove unwanted frequencies from a signal.

Filter Classification – Amplitude Response

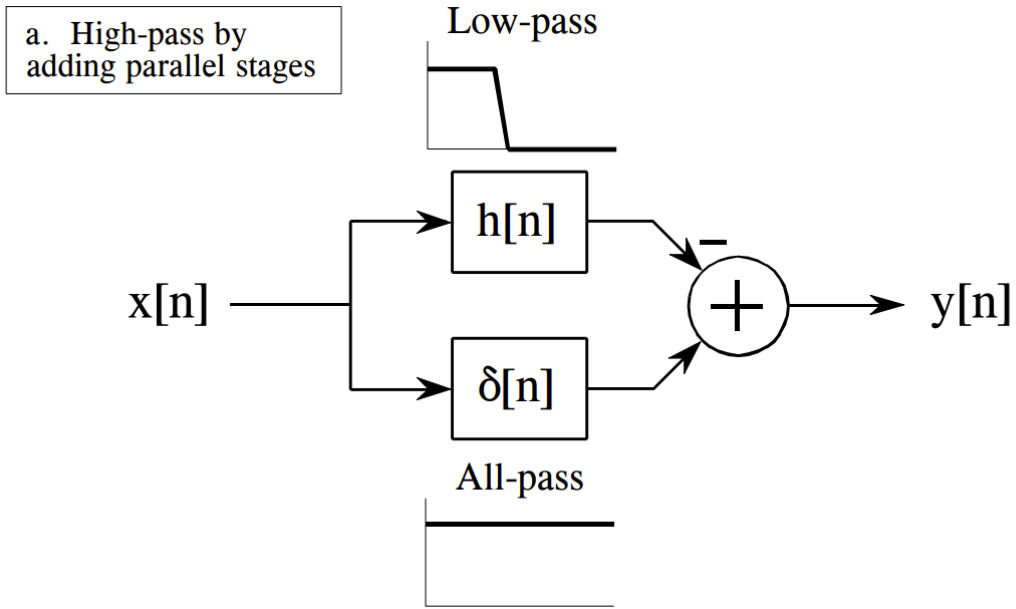
Band-stop filter

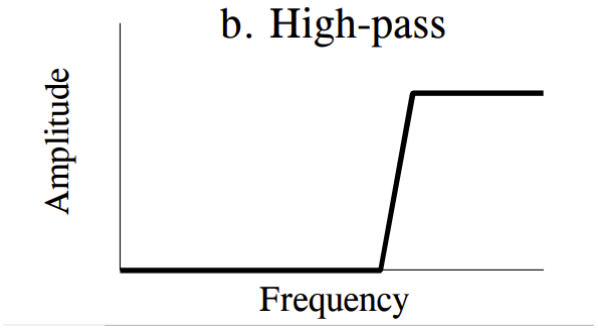


- A band-stop filter attenuates a range of frequencies while allowing frequencies outside of that range to pass through.
- It is characterized by a stop-band that includes a range of frequencies and two pass-bands on either side of the passband
- A band-stop filter is often used to remove unwanted frequency ranges from a signal.

Filter Classification – Amplitude Response

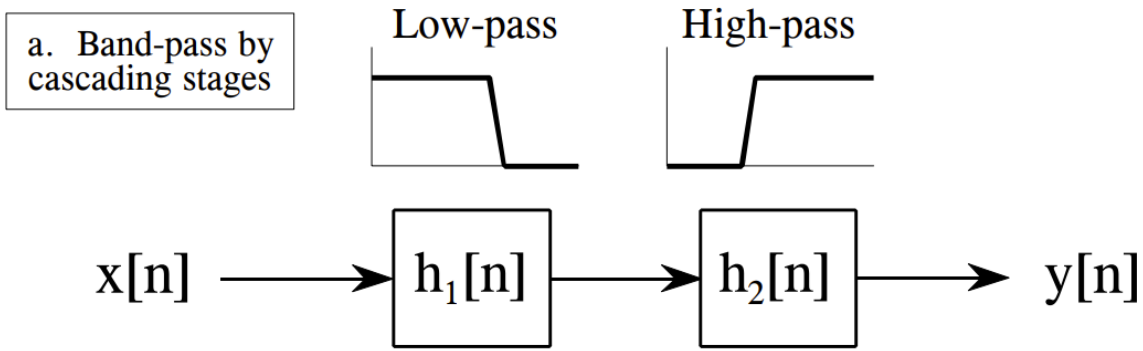
Convert Lowpass to High-pass

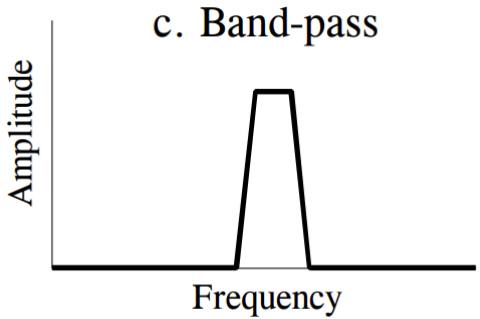




Filter Classification – Amplitude Response

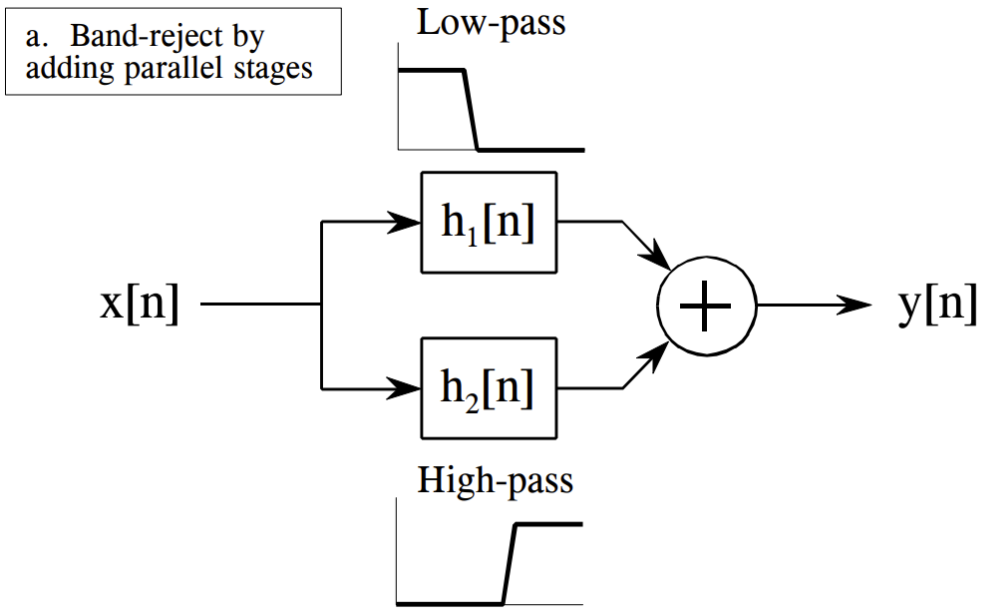
Convert Lowpass to Band-pass

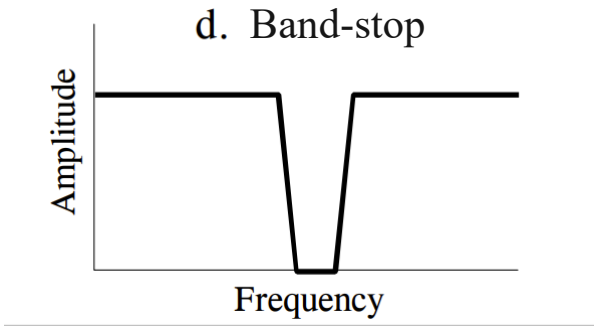




Filter Classification – Amplitude Response

Convert Lowpass to Band-stop



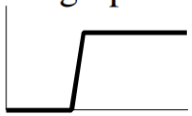


Filter Classification – Amplitude Response

Convert Lowpass to …

b. High-pass
in a single stage

High-pass



$x[n] \longrightarrow \boxed{\delta[n] - h[n]} \longrightarrow y[n]$

b. Band-pass
in a single stage

Band-pass

$x[n] \longrightarrow \boxed{h_1[n] * h_2[n]} \longrightarrow y[n]$

b. Band-reject
in a single stage

Band-stop

$x[n] \longrightarrow \boxed{h_1[n] + h_2[n]} \longrightarrow y[n]$

Filter Example

<u>Example</u>: Consider this filter $y[n] - 0.5y[n-1] = x[n]$

1. Find the impulse response when $y[-1] = 0$
2. Derive the transfer function
3. Determine the stability of this filter
4. Compute the amplitude response
5. Evaluate the amplitude response for frequencies $\omega = [0, 0.2\pi, 0.4\pi, 0.6\pi, 0.8\pi, \pi]$
6. State whether this is a low-pass / band-pass / high-pass / band-stop
7. State whether this is a FIR filter or a IIR filter.

Filter Example

<u>Example</u>: Consider this filter $y[n] - 0.5y[n-1] = x[n]$

1. Find the impulse response when $y[-1] = 0$

Need to solve the difference equation when $x[n] = \delta[n]$.

The solution is $y[n] = (0.5)^n u[n]$

Filter Example

<u>Example</u>: Consider this filter $y[n] - 0.5y[n-1] = x[n]$

2. Derive the transfer function

$$H(z) = \frac{X(z)}{Y(z)} = \frac{1}{1 - 0.5z^{-1}}$$

Filter Example

<u>Example</u>: Consider this filter $y[n] - 0.5y[n-1] = x[n]$

3. Determine the stability of this filter

Since $H(z) = \frac{1}{1-0.5z^{-1}}$, the filter has only one pole at $z = 0.5$

This pole is inside the unit circle, hence the filter is stable

Filter Example

<u>Example</u>: Consider this filter $y[n] - 0.5y[n-1] = x[n]$

4. Compute the amplitude response

Based on $H(z) = \frac{1}{1-0.5z^{-1}}$, the frequency response is given by

$$H\left(e^{j\omega}\right) = \frac{1}{1 - 0.5e^{-j\omega}}$$

Hence, the amplitude response is

$$\left|H\left(e^{j\omega}\right)\right| = \frac{1}{|1 - 0.5e^{-j\omega}|} = \frac{1}{\sqrt{(1 - 0.5e^{-j\omega})(1 - 0.5e^{j\omega})}}$$

$$= \frac{1}{\sqrt{1.25 - \cos\omega}}$$

Filter Example

<u>Example</u>: Consider this filter $y[n] - 0.5y[n-1] = x[n]$

5. Evaluate the amplitude response for frequencies $\omega = [0, 0.2\pi, 0.4\pi, 0.6\pi, 0.8\pi, \pi]$

By substituting the values of ω to $\left|H\left(e^{j\omega}\right)\right| = \frac{1}{\sqrt{1.25 - \cos\omega}}$

We get the following amplitude responses

$$[2.0000 \quad 1.5059 \quad 1.0309 \quad 0.8009 \quad 0.6969 \quad 0.6667]$$

Filter Example

<u>Example</u>: Consider this filter $y[n] - 0.5y[n-1] = x[n]$

6. State whether this is a low-pass / band-pass / high-pass / band-stop

The amplitude response is

$$[2.0000 \quad 1.5059 \quad 1.0309 \quad 0.8009 \quad 0.6969 \quad 0.6667]$$

The amplitude response decreases as the (normalized) frequency increases from 0 to π

Hence, this is a low-pass filter

Filter Example

Example: Consider this filter $y[n] - 0.5y[n-1] = x[n]$

7. State whether this is a FIR filter or a IIR filter.

By looking at the impulse response $y[n] = (0.5)^n u[n]$, this function is non-zero for all values of n. Hence, it is an IIR filter.

Equivalently:
By looking at the difference equation, the coefficients of past output $y[n-1]$ is -0.5. So the difference equation is recursive. Hence, it can only be implemented by recursion, which means it is an IIR filter

Lecture summary

- **Frequency Response** shows how the filter respond to different frequencies. Its amplitude has three parameters:
  1. Passband
  2. Transition band
  3. Stopband
- Impulse, step, and frequency responses all contain identical information, represented in different ways.
- Both impulse response and step response are **time-domain response**, where frequency response is a **frequency-domain response**

Lecture summary

- Filters are classified in two ways:
  - The amplitude response classify filters into
    - ✓ Low-pass filters
    - ✓ High-pass filters
    - ✓ Band-pass filters
    - ✓ Band-stop filters
  - The implementation method classifies filters into
    - ✓ FIR filters (implemented by convolution)
    - ✓ IIR filters (implemented by recursive equations)

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